

Qwen_VL Streamlit Colab Project

Visual Question Answering (VQA) System

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Powered by Qwen2-VL-2B-Instruct

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Qwen_VL Streamlit Colab App Project Documentation

Overview

This project demonstrates a **visual-language AI system** built using **Qwen2-VL-2B-Instruct**, an open-source multimodal model by Alibaba Cloud.

It can understand **both images and text**, making it capable of **Visual Question Answering (VQA)**, **caption generation**, and **image-based reasoning**.

The system is deployed via **Streamlit** and hosted publicly through **Cloudflare Tunnel** inside **Google Colab**, enabling anyone to access the model through a web interface.

System Architecture

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Google Colab Notebook
(Python Runtime)

Streamlit Web App

Qwen2-VL-2B-Instruct
(HuggingFace Transformers)

Cloudflare Tunnel (Public URL)

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Accessible Web UI

...

Key Components

Component	Description
Model	`Qwen2-VL-2B-Instruct` a multimodal model supporting vision-language tasks.
Framework	Streamlit builds the interactive frontend interface.
Deployment	Cloudflared tunnel creates a secure, sharable public link from Colab.
Backend Libraries	`transformers`, `torch`, `accelerate`, `pillow`.
Environment	Google Colab (Python 3.10+, GPU runtime recommended).

Features

- Visual Question Answering (VQA)
- OCR + QA Mode
- Multimodal Reasoning
- Web Interface via Cloudflare Tunnel
- Colab Deployable

Setup Instructions

```
bash
!pip install -q streamlit cloudflared transformers accelerate pillow torch
!nohup streamlit run app.py --server.port 8501 > /dev/null 2>&1 &
import time; time.sleep(5)
!cloudflared tunnel --url http://localhost:8501 --no-autoupdate
...
```

Streamlit App Structure

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```
```python
import streamlit as st

from transformers import AutoProcessor, AutoModelForVision2Seq
from PIL import Image
import torch

model_name = "Qwen/Qwen2-VL-2B-Instruct"
processor = AutoProcessor.from_pretrained(model_name)
model = AutoModelForVision2Seq.from_pretrained(model_name, torch_dtype=torch.float16,
device_map="auto")

st.title(" Qwen-VL Image Understanding App")
uploaded_image = st.file_uploader("Upload an image", type=["jpg", "png", "jpeg"])
question = st.text_input("Ask a question about the image:")

if uploaded_image and question:
 image = Image.open(uploaded_image)
 inputs = processor(text=question, images=image, return_tensors="pt").to("cuda")
 output = model.generate(**inputs, max_new_tokens=128)
 answer = processor.decode(output[0], skip_special_tokens=True)
 st.image(image, caption="Uploaded Image", use_column_width=True)
 st.markdown(f"Answer: {answer}")
```
```

Troubleshooting

| Problem | Cause | Fix |

|-----|-----|-----|

| ****DNS_PROBE_FINISHED_NXDOMAIN**** | Cloudflare tunnel failed | Use `--no-autoupdate`` and ensure Streamlit is running first. |

| ****Black screen / no output**** | Streamlit crashed | Run `!streamlit run app.py`` to check logs. |

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| **GPU not used** | Colab CPU runtime | Switch to GPU under **Runtime** Change runtime type GPU*. |

Credits

- **Model:** [Qwen2-VL-2B-Instruct](https://huggingface.co/Qwen/Qwen2-VL-2B-Instruct)
- **Libraries:** Hugging Face Transformers, PyTorch, Streamlit, Pillow
- **Author:** Hemanth V
- **Platform:** Google Colab (with Cloudflare Tunnel)

Future Improvements

- Add **speech input/output** using `gTTS` or `pyttsx3`.
- Implement **multi-image comparison**.
- Enhance **handwritten text recognition** using hybrid OCR models.
- Deploy permanently on **Streamlit Cloud / Hugging Face Spaces**.